Theta manifold geometry and signed heat-flow collision control

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Abstract

We study the collision detector $E = H^2 + H_x^2$ for the complete theta heat flow through signed jet geometry and arithmetic source identities. Every finite collision forces a negative two-point chord of order $\sqrt{E}$ at its stationary predecessor and a fixed nearby endpoint. Consequently a source-prescribed lower bound with error $o(\sqrt{E})$, uniform on the required short intervals, excludes all finite collision multiplicities. The exponent $1/2$ is sharp for arbitrary power-law allowances. An all-cutoff estimate transfers to this criterion even with Gaussian cutoff growth when its rate is strictly below the tail rate. We give complete signed correlation formulas, including primitive endpoints and transverse theta-jet commutators, in an explicit curved elliptic-family geometry. A validated weighted-square collision shows that support positivity and the first source endpoint are insufficient; full modular reflection instead forces the theta coefficients, without by itself proving the needed sign. Holomorphic jet surfaces and their real Gauss maps explain the collision concentration. The remaining problem is a theta-specific chord estimate with full domain coverage. No new global theta sign bound or proof of RH is claimed.

Contents

1 The geometric problem and the program **2**
1.1 Results and their logical status . . . 3
1.2 Relation to the preceding investigation . . . 3
1.3 Conventions . . . 4

2 The physical heat field and its collision obstruction **4**
2.1 Full stationary predecessors at every finite multiplicity . . . 5

3 Signed forms and the physical observation **7**
3.1 The Hodge pairing in four dimensions . . . 7
3.2 Primitives and a signed boundary observation . . . 8
3.3 An integrated positive identity and its scale . . . 8

4 Hyperkähler lifts and their degenerations **9**
4.1 A flat lift with an exact physical coefficient . . . 9
4.2 Conformal and compact obstructions . . . 10

5 A holomorphic surface with the physical metric coefficient **10**
5.1 Curvature and induced holonomy . . . 11
5.2 Genuine calibrated inclusions and their limits . . . 12

6 The marked real Gauss map retains the sign **12**
6.1 A sign-reversing geometric control 13

7 Conditional exclusion by signed chords and spatial turning **14**
7.1 A fixed nearby endpoint and starting-point energy 14
7.2 Transfer from a complete family of cutoff estimates 16
7.3 The inner profile and quarter-turn concentration 17
7.4 A supporting one-sided continuity criterion 18

8 The complete theta source on a curved elliptic family **18**
8.1 The complete primitive and its boundary contribution 19
8.2 The theta line bundle, its connection, and the fourth jet 20
8.3 A nonflat hyperkähler metric with the correct elliptic fibers 22
8.4 An actual abelian anti-self-dual theta connection 23
8.5 Physical restriction and complete signed source identities 24

9 Arithmetic discrimination: support, endpoints and reflection **26**
9.1 A validated collision within the square-support relaxation 26
9.2 Complete reflection fixes the coefficients 28

10 The remaining arithmetic-to-physical estimate **28**
10.1 Coefficient coupling with a complete remainder 29
10.2 A tested contour route and its limitation 29
10.3 Validation and coverage 29

A Verification scope and provenance **30**

1 The geometric problem and the program

The theta heat flow presents a tension between positive source data and a signed spectral observation. The complete source is positive and has very rapid decay, but its oscillatory transform can have small first jets. The physical collision detector is

$$E(t,x) = H(t,x)^2 + H_x(t,x)^2.$$

The research objective is to exclude $E = 0$ at finite positive heat time on the full real spectral line. The established de Bruijn–Newman framework supplies the relation to RH, stated with its domain requirements below. This paper develops the differential geometry of that objective; it does not prove the missing theta-specific estimate.

There are two natural manifolds. On the physical side, a primitive $P_z = H$ gives a holomorphic surface $X_t(z,w) = (P(t,z), H(t,z), w)$ whose induced metric contains the complex first-jet energy as a coefficient. Its regular locus is a Kähler manifold; a common zero of H and H_z is a branch event for the parametrization. The real projective tangent direction retains the signed Laguerre quantity through

$$\partial_x \arg(H + iH_x) = -\frac{H_x^2 - HH_{xx}}{E}.$$

The associated two-point observation is

$$\mathcal{C}_j(t;x,y) = h_j(t,x)\{h_j(t,y) - h_{j+1}(t,y)\}, \qquad h_j = \partial_x^j H, \quad j = 0,1.$$

At a full stationary start $h_{j+1}(t,x) = 0$, on an interval with $h_j \neq 0$, its nonnegativity is precisely a $\pi/4$ upper bound on the rightward projective turning. More usefully, every collision forces $\mathcal{C}_j/\sqrt{E(t,x)}$ to approach a strictly negative constant at a fixed nearby endpoint. A lower bound with error $o(\sqrt{E(t,x)})$ therefore suffices across all finite multiplicities. The energy is evaluated at the starting point on both parity branches.

On the source side, the actual theta kernel is a covariant derivative expression in a degree-one line bundle over a curved elliptic family. The complete two-source transform retains its spectral phases, positive heat multiplier, transverse jet corrections and primitive endpoints. Its directly restricted connection is flat, so curvature alone adds no new tangential sign. The arithmetic question is sharper: which coupling of the theta coefficients can control this complete signed observation? A positive weighted-square source can collide while retaining the first source endpoint; full modular reflection excludes those free weights by forcing the theta coefficients. Turning this rigidity into an inequality remains open.

1.1 Results and their logical status

The principal results and the remaining candidate are as follows.

1. An all-multiplicity fixed-endpoint chord theorem gives conditional collision exclusion from an $o(\sqrt{E})$ starting-energy error. The threshold exponent is sharp. The inner and parabolic heat profiles explain the same positive turning concentration at shrinking scales.

2. A uniform all-cutoff estimate with cost $\exp(aN^{\beta})$ and Gaussian tail transfers to exclusion for $\beta < 2$. For the chord, $\beta = 2$ also works under a strict local gap between the growth and tail rates, with the integer-rounding loss retained.

3. Exact complete-source formulas express the chord and stationary Laguerre observations in both covariant theta jets and primitive variables. The affine endpoint terms and the distinction between flat path restriction and transverse differentiation are explicit.

4. A validated non-theta weighted-square collision rejects support positivity and the first endpoint as sufficient hypotheses. An analytic reflection-rigidity theorem identifies a stronger coefficient coupling that any successful arithmetic mechanism may use.

5. Signed Hodge identities, holomorphic jet surfaces and calibrated inclusions give the geometric setting. A nonflat semi-flat hyperkähler metric and an abelian ASD theta connection are constructed; the latter's base metric freedom removes its displayed physical-path growth without producing the missing sign estimate.

The strongest prospective route is a complete signed chord bound using coefficient coupling, with a remainder paid by the cutoff-transfer theorem. No such theta bound is proved here. The analytic exclusion theorems are conditional; the weighted-square result concerns a control source; the finite checks validate specified calculations. These roles must not be conflated with a new theta sign domain or a proof of RH. We make no novelty claim for the standard geometric constructions.

1.2 Relation to the preceding investigation

This investigation separates the manifold and connection direction from the broader signed-theta program in [1]. The preceding manuscript develops complete-source correlations, cutoff accounting, outer-region estimates and finite certificates. Its self-dual construction and its later manifold notes are the starting point of the present narrative. The outer estimates and certificates retain their

original statements and verification records; they are not reproved or used as a replacement for the missing global geometric estimate here.

The author's exploratory quaternion-form notes of January and February 2014 [2] motivate preserving a signed, oriented pairing. We build the required identities directly and state their boundary hypotheses. The new line-bundle construction is abelian and has an explicit connection and curvature; it invokes no unproved general nonabelian boundary formula.

Throughout, a regular jet interval means an interval on which the particular pair defining the angle is nowhere zero. Source-prescribed constants or moduli must be derived from the complete theta data and remain uniform through neighborhoods of hypothetical collisions. A constant chosen retrospectively from an observed positive energy floor does not provide such an estimate.

1.3 Conventions

All metrics are real Riemannian metrics. We use $|dz|^2 = dx^2 + dy^2$ and the Kähler form $(i/2)\, dz \wedge d\bar{z}$. The physical time t is an external parameter for the four-dimensional exterior derivatives. It is distinct from the elliptic modulus τ, whose imaginary part is positive. Hodge signs refer to the orientation stated in each construction. For a Hermitian holomorphic line bundle in a holomorphic frame with squared norm h, we use the Chern connection $\nabla = d + \partial \log h$ and curvature $F_\nabla = \overline{\partial}\partial \log h$; a frame change transforms the local connection in the usual way.

2 The physical heat field and its collision obstruction

The geometric constructions in this paper must retain a specified physical observation. We first fix that observation and prove the local analytic facts used later. The conclusions in this section about arbitrary analytic heat fields are local theorems. Their implication for the Riemann hypothesis uses an explicitly identified earlier theorem for the complete theta source.

For real $u \geq 0$, define

$$\Phi(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \tag{2.1}$$

$$H(t,z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du. \tag{2.2}$$

The variable t is an external heat parameter. In the geometry below it is fixed when spatial derivatives, metrics, and curvature are computed. On real (t,x) put

$$h_j = \partial_x^j H, \qquad E = h_0^2 + h_1^2, \qquad \mathcal{L}_j = h_{j+1}^2 - h_j h_{j+2}, \qquad S_j = \{h_{j+1} = 0\}, \quad j = 0, 1. \tag{2.3}$$

The same notation $h_j(t,z)$ denotes the holomorphic spectral derivatives. The four consecutive components h_0, h_1, h_2, h_3 have one scalar source; they are not independent fields to which arbitrary geometric constraints may be assigned.

Lemma 2.1 (Analyticity and the sign of heat time)**.** *The integral (2.2) defines a jointly entire function of $(t,z) \in \mathbb{C}^2$. It is even in z, obeys real reflection for real t, and satisfies*

$$H_t = -H_{zz}. \tag{2.4}$$

For every real t, $H(t,0) > 0$. Consequently no real-time slice is identically zero and every spatial zero has finite multiplicity.

Proof. Every term of (2.1) is positive for $u \geq 0$, since $2\pi n^2 e^{4u} - 3 > 0$. Dropping the negative term and comparing successive terms of $n^4 e^{-\pi n^2 e^{4u}}$ gives

$$0 < \Phi(u) < 4\pi^2 e^{9u - \pi e^{4u}}.$$

Indeed the ratio of successive majorant terms is at most $16e^{-3\pi} < 1/2$. On each compact set of complex (t, z), every fixed parameter derivative of the integrand is bounded by a constant times

$$(1+u)^M \exp\{Cu^2 + Cu + 9u - \pi e^{4u}\},$$

which is integrable. Uniform convergence and differentiation under the integral prove joint entirety and (2.4). The symmetries follow from the cosine representation. At real t and $z = 0$ the integrand is strictly positive. Finally, zeros of a nonzero holomorphic function have finite order. □

Direct differentiation, with x held fixed, gives

$$\frac{1}{2}E_t = -h_0 h_2 - h_1 h_3 = \begin{cases} \mathcal{L}_0, & h_1 = 0, \\ \mathcal{L}_1, & h_2 = 0. \end{cases} \tag{2.5}$$

These are the stationary sets of the original field and its first derivative. A normalization of the field depending on x would change its stationary set and cannot be substituted in this identity.

Remark 2.2 (The imported theta-to-RH bridge). For the normalization (2.1)–(2.2), the earlier investigation establishes

$$H(0, z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right)$$

and the finite-collision criterion

$$\text{RH} \quad \Longleftrightarrow \quad E(t, x) > 0 \quad (0 < t \leq 1/2,\ x \in \mathbb{R}). \tag{2.6}$$

We use this result as an imported theorem, with its normalization and range unchanged; see [1, section on positive-time collisions]. Its global inputs are the de Bruijn–Newman real-zero threshold, including its endpoint and upper bound $1/2$, the strip bound for nonreal zeros, and a confinement estimate uniform for t in compact positive-time intervals. The last input prevents a limiting first collision from escaping to infinite real height. Joint analyticity and conjugation symmetry then force a finite multiple real zero if RH fails. Conversely, local backward-heat splitting at a positive-time multiple zero produces nonreal zeros immediately before it. Thus a geometric construction alone does not supply (2.6); its global theta hypotheses remain part of the imported theorem. The local exclusion arguments below are proved independently of that import.

2.1 Full stationary predecessors at every finite multiplicity

Lemma 2.3 (Even-order stationary predecessor)**.** *Let $W(t, x)$ be real analytic near (T, a), satisfy $W_t = -W_{xx}$, and have an exact spatial zero of order $2r$ there, where $r \geq 1$. Write $A = \partial_x^{2r} W(T, a) \neq 0$ and $B = \partial_x^{2r+1} W(T, a)$. For sufficiently small $s \geq 0$ there is a real analytic curve $\chi(s)$ such that*

$$W_x(T - s, \chi(s)) = 0, \qquad \chi(s) = a - \frac{B}{rA}s + O(s^2), \tag{2.7}$$

$$W(T - s, \chi(s)) = \frac{A}{r!}s^r + O(s^{r+1}), \qquad W_{xx}(T - s, \chi(s)) = \frac{A}{(r-1)!}s^{r-1} + O(s^r). \tag{2.8}$$

In particular the central value of W is nonzero for sufficiently small $s > 0$.

Proof. Analyticity and the heat equation give the convergent Taylor expansion

$$W(T-s,a+y)=\sum_{p,q\geq 0}\frac{s^p y^q}{p!q!}\,\partial_x^{2p+q}W(T,a). \tag{2.9}$$

Set $y=s\eta$ and differentiate once in y. Every surviving term has $2p+q+1\geq 2r$, hence $p+q\geq r$. It follows that $s^{-r}W_x(T-s,a+s\eta)$ extends analytically across $s=0$. At $s=0$ its value is

$$\frac{A}{(r-1)!}\eta+\frac{B}{r!}.$$

Its derivative in η is nonzero. The analytic implicit-function theorem supplies $\eta(s)$ with initial value $-B/(rA)$, proving (2.7). Substitution in (2.9) proves (2.8): the leading terms are respectively $p=r,q=0$ and, after two spatial derivatives, $p=r-1,q=0$. □

Proposition 2.4 (The physical stationary pole)**.** *Suppose a nonzero analytic backward-heat field H has exact spatial multiplicity $m=2r+j$ at (T,a), with $r\geq 1$ and $j\in\{0,1\}$. Apply Lemma 2.3 to $W=h_j$. On its full stationary predecessor, as $s\downarrow 0$,*

$$h_{j+1}=0,\qquad E>0,\qquad \frac{\mathcal{L}_j}{E}=-\frac{r}{s}+O(1). \tag{2.10}$$

For $j=0$ one has $E=W^2$ exactly. For $j=1$,

$$H(T-s,\chi(s))=O(s^{r+1}),\qquad E=W^2(1+O(s^2)). \tag{2.11}$$

Proof. On the constructed branch $\mathcal{L}_j=-WW_{xx}$. Equation (2.8) gives $W_{xx}/W=r/s+O(1)$. If $j=0$, stationarity is $H_x=0$, so $E=W^2$ and the claim follows. If $j=1$, H has order $2r+1$ at the event. In its heat Taylor expansion evaluated at $x-a=O(s)$, each surviving term has $2p+q\geq 2r+1$, which implies $p+q\geq r+1$. Thus $H=O(s^{r+1})$, whereas W is a nonzero constant times s^r. This proves (2.11) and then (2.10). All divisions are taken on the proved nonzero predecessor, not at the event itself. □

Theorem 2.5 (Signed stationary exclusion)**.** *Let H be a real analytic backward-heat field whose spatial slices are not identically zero. Suppose nonnegative functions K_0,K_1 are locally bounded through every candidate event in a stated time-space domain and*

$$\mathcal{L}_j\geq -K_jE\quad \text{on the full set } S_j,\qquad j=0,1. \tag{2.12}$$

Assume the domain contains the sufficiently close predecessors of each candidate event. Then it contains no multiple real zero covered in this way. For the complete theta field, if the hypothesis covers $(0,1/2]\times\mathbb{R}$, it implies RH by Remark 2.2.

Proof. Every multiple zero has a finite order $m\geq 2$, uniquely expressible as $2r+j$ with the stated ranges. Proposition 2.4 then contradicts the locally bounded coefficient K_j on its corresponding predecessor. At the endpoint $t=1/2$, a left-hand time neighborhood suffices. No boundedness as $t\downarrow 0$ or as $|x|\to\infty$ is required. □

Both branches are essential for this argument. The backward-heat polynomial $H(t,x)=x^3-6(t-T)x$ has $\mathcal{L}_0=3x^4+36(t-T)^2\geq 0$ everywhere and nevertheless has a triple zero at $(T,0)$. On the second stationary predecessor $x=0$, $\mathcal{L}_1/E=-1/(T-t)$.

3 Signed forms and the physical observation

The two Laguerre expressions are polarized quadratic forms in three consecutive derivatives. A geometric representation must preserve that polarization. Replacing it by the norm of the whole jet changes the question, because the higher derivatives need not vanish when E does. The quaternion-form working notes of Edward B. Baker III [2] motivate the use of oriented two-form pairings. The construction below is derived directly; it uses none of the unfinished singular-kernel or nonabelian boundary formulas in those notes.

3.1 The Hodge pairing in four dimensions

Fix the Euclidean metric on $\mathbb{R}^4$ with coordinates (x, y, a, b), orientation $dx \wedge dy \wedge da \wedge db$, and complex coordinates $z = x + iy$, $w = a + ib$. This use of a, b avoids confusing the auxiliary manifold coordinates with the source variable u. The Hodge star gives

$$\Lambda^2 = \Lambda^2_+ \oplus \Lambda^2_-, \qquad \alpha \wedge \alpha = (|\alpha_+|^2 - |\alpha_-|^2)\, dV.$$

The wedge pairing has signature $(3, 3)$; the inner product $\alpha \wedge *\alpha$ is positive definite. Both structures are useful, but they answer different questions.

For orthonormal forms $\sigma_1^+, \sigma_2^+ \in \Lambda^2_+$ and $\sigma_1^- \in \Lambda^2_-$, the form

$$\Omega = B\sigma_1^+ + \tfrac{1}{2}(A - C)\sigma_2^+ + \tfrac{1}{2}(A + C)\sigma_1^-$$

has wedge square $(B^2 - AC)dV$ and squared norm $B^2 + (A^2 + C^2)/2$. This realizes a Laguerre expression algebraically, but its norm pays for the higher jet C, and a variable-coefficient version need not be closed. We instead realize each input by a closed form and retain the signed polarization explicitly.

Proposition 3.1 (Closed self-dual inputs)**.** *For a holomorphic function $f(z)$, define*

$$\alpha_f = \frac{1}{\sqrt{2}} \operatorname{Re}(f(z)\, dz \wedge dw).$$

Then

$$d\alpha_f = 0, \qquad *\alpha_f = \alpha_f, \qquad \alpha_f \wedge \alpha_g = \operatorname{Re}(f\overline{g})\, dV. \tag{3.1}$$

The form is Hodge-harmonic, and in the fixed Euclidean metric

$$|\alpha_f|^2 = |f|^2, \qquad |\nabla \alpha_f|^2 = 2|f'|^2. \tag{3.2}$$

In particular, it is parallel on a connected domain exactly when f is constant.

Proof. Put $A = dx \wedge da - dy \wedge db$ and $B = dx \wedge db + dy \wedge da$. Both are self-dual, $A^2 = B^2 = 2dV$, and $A \wedge B = 0$. If $f = p + iq$, then $\alpha_f = (pA - qB)/\sqrt{2}$, which proves the pairing and norm. Holomorphicity gives $d(f\, dz \wedge dw) = 0$. Self-duality then implies $d^*\alpha_f = - * d * \alpha_f = 0$. The Cauchy–Riemann equations give $|\nabla p|^2 + |\nabla q|^2 = 2|f'|^2$, proving the last identity. □

For $f = h_j(t, \cdot)$, heat time is held fixed in this exterior derivative. Writing $\alpha_j = \alpha_{h_j}$, we obtain

$$\alpha_{j+1} \wedge \alpha_{j+1} - \alpha_j \wedge \alpha_{j+2} = \big(|h_{j+1}|^2 - \operatorname{Re}(h_j \overline{h_{j+2}})\big) dV. \tag{3.3}$$

On $y = 0$ the coefficient is $\mathcal{L}_j$. This is evaluation of a four-form coefficient on the real slice, not pullback of a four-form onto a one-dimensional line. The coefficient-space matrix has spectrum $\{1, 1/2, -1/2\}$. Its negative eigenvalue persists even though all three input forms are closed and self-dual.

3.2 Primitives and a signed boundary observation

If $P_f' = f$, then

$$\beta_f = \frac{1}{\sqrt{2}} \operatorname{Re}(P_f\, dw), \qquad d\beta_f = \alpha_f.$$

Stokes' theorem gives, for a smooth bounded four-domain with the forms defined on a neighborhood of its closure,

$$\int_\Omega \alpha_f \wedge \alpha_g = \int_{\partial\Omega} \beta_f \wedge \alpha_g. \tag{3.4}$$

This retains a mixed signed pairing. It does not replace the boundary term by a positive norm or remove it.

The following elementary identity makes the same distinction on a spectral circle. Write $\langle q \rangle = (2\pi)^{-1} \int_0^{2\pi} q(\theta)\, d\theta$.

Proposition 3.2 (Complete circle current)**.** *Let f be holomorphic near the closed disk $|z - x_0| \le R$, real on its real diameter, and $f'(x_0) = 0$. Set*

$$\varphi(\theta) = f(x_0) \operatorname{Re} f(x_0 + Re^{i\theta}).$$

Then

$$-f(x_0) f''(x_0) = -\frac{4 f(x_0)^2}{R^2} + \frac{4}{R^2} \langle (1 - \cos 2\theta) \varphi(\theta) \rangle. \tag{3.5}$$

For $f = h_j$ this is exactly $\mathcal{L}_j$ on its full stationary set.

Proof. The Taylor coefficients are real. The mean of $\operatorname{Re} f$ is $f(x_0)$ and its second cosine coefficient has mean pairing $\langle \cos 2\theta \operatorname{Re} f \rangle = R^2 f''(x_0)/4$. Substitution proves the formula. □

The current φ is also the coefficient of $\alpha_{f(x_0)} \wedge \alpha_f$ evaluated on the circle. A useful bound must control the complete weighted current relative to E, with all radius costs accounted for. Separately estimating its negative mass can discard essential favorable cancellation.

3.3 An integrated positive identity and its scale

The following result, inherited from the signed-form development in [1], explains what positivity of a completed integral can provide without supplying the central pointwise estimate.

Theorem 3.3 (Disk completion)**.** *For f holomorphic near $\overline{D_R(x_0)}$, define*

$$\ell_f = |f'|^2 - \operatorname{Re}(f\overline{f''}), \qquad g_R(z) = \left(1 - \frac{(z - x_0)^2}{R^2}\right) f(z).$$

Then

$$\int_{D_R} \ell_f\, dA + \frac{1}{2R} \int_{\partial D_R} |f|^2\, ds = \frac{1}{2} \int_{D_R} |g_R'|^2\, dA \ge 0. \tag{3.6}$$

The boundary coefficient cannot be decreased, even for real-reflection polynomials stationary at the center.

Proof. Write $f(x_0 + R\zeta) = \sum_{n \ge 0} b_n \zeta^n$ and set $b_n = 0$ for $n < 0$. Angular orthogonality gives

$$\pi^{-1} \int_{D_R} \ell_f\, dA = \sum_{n \ge 0} n |b_n|^2 - \operatorname{Re} \sum_{n \ge 0} (n + 2) b_n \overline{b_{n+2}},$$

$$(2\pi R)^{-1}\int_{\partial D_R}|f|^2\,ds=\sum_{n\ge 0}|b_n|^2,$$
$$\pi^{-1}\int_{D_R}|g_R'|^2\,dA=\sum_{n\ge 1}n|b_n-b_{n-2}|^2.$$

Expansion proves (3.6). For $f_N=\sum_{m=0}^{N}(1-m/(N+1))((z-x_0)/R)^{2m}$, direct summation gives

$$B_N:=\frac{1}{2R}\int_{\partial D_R}|f_N|^2\,ds=\pi\frac{(N+2)(2N+3)}{6(N+1)},\qquad \frac{\int_{D_R}\ell_{f_N}\,dA}{B_N}=-\frac{2N}{2N+3}\longrightarrow -1.$$

These polynomials have $f_N(x_0)=1$ and $f_N'(x_0)=0$. □

Their boundary energy grows while their central first-jet energy stays equal to one. Thus the integrated positive completion cannot be converted to a uniform central-energy payment for arbitrary holomorphic fields. The actual theta source must supply the additional restriction.

4 Hyperkähler lifts and their degenerations

A hyperkähler metric in real dimension four has a parallel compatible triple of complex structures satisfying the quaternion relations; its holonomy is contained in $\mathrm{Sp}(1)=\mathrm{SU}(2)$. In particular it is Ricci-flat. These are statements about the metric and its parallel tensors, not a consequence of closed self-duality of one form; see [4].

4.1 A flat lift with an exact physical coefficient

Proposition 4.1 (Pullback hyperkähler metric)**.** *Where a holomorphic $f(z)$ is nonzero, the metric*

$$g_f=|f(z)|^2|dz|^2+|dw|^2 \tag{4.1}$$

is locally flat and hyperkähler. Its parallel triple is

$$\omega_1=\frac{i}{2}(|f|^2dz\wedge d\overline{z}+dw\wedge d\overline{w}),$$
$$\omega_2=\mathrm{Re}(f\,dz\wedge dw),\qquad \omega_3=\mathrm{Im}(f\,dz\wedge dw).$$

For $f=H+iH_z$, its real-slice coefficient is exactly E.

Proof. A local primitive $F'=f$ supplies the nonsingular holomorphic coordinate $\zeta=F(z)$. Then $g_f=|d\zeta|^2+|dw|^2$ and the displayed forms are the standard constant hyperkähler triple in those coordinates. Equivalently they are closed and $\omega_i\wedge\omega_j=2\delta_{ij}|f|^2dV$. For real x, both H and H_x are real, so $|H+iH_x|^2=H^2+H_x^2$. □

This construction has trivial restricted holonomy, contained in $\mathrm{Sp}(1)$; it does not have holonomy equal to the full group. It is a precise geometric realization of the physical scale. At a zero of f of order m, however, $F-F(z_0)$ has order $m+1$ and (4.1) degenerates. After a nonsingular holomorphic change of coordinate its curve factor is a constant times $|\zeta|^{2m}|d\zeta|^2$. Its metric completion has cone angle $2\pi(m+1)$ and finite distance to the vertex. Passing to the branched coordinate F is not a nonsingular coordinate repair on the original domain.

At a nondegenerate physical collision $H=H_z=0$, $H_{zz}\neq 0$, the coefficient $f=H+iH_z$ has a simple zero since $f'=iH_{zz}$. Thus the real physical collision produces an angle-4π degeneration. Away from the real slice, $H+iH_z$ can vanish without both jets vanishing; the exact equivalence with the physical detector is a real-slice statement. The fixed background metric remains smooth at every such event, but its theta-dependent forms are generally harmonic rather than parallel.

4.2 Conformal and compact obstructions

In dimension four the Hodge star on two-forms is unchanged by $g \mapsto e^{2\sigma} g$. Thus closed self-dual forms remain harmonic under arbitrary conformal changes. For example, $e^{2x} g_0$ has

$$\mathrm{Ric} = -2(dy^2 + da^2 + db^2), \qquad \mathrm{Scal} = -6e^{-2x}.$$

It is not hyperkähler, despite preserving all the closed self-dual inputs. The top-form pairing itself is unchanged; only its coefficient relative to the volume changes, by a positive factor $e^{-4\sigma}$. No sign improvement follows from this conformal replacement.

Pointwise normalization also needs care. Where $f \neq 0$,

$$d(\alpha_f/|f|) = -d\log|f| \wedge (\alpha_f/|f|).$$

A nonzero self-dual two-form is symplectic, so wedge multiplication by it is an isomorphism from one-forms to three-forms. Consequently this normalized form is closed only if $|f|$ is constant, which for holomorphic f forces f constant. Normalization can therefore lose the differential equation needed for a holonomy construction.

Proposition 4.2 (Compact exactness obstruction)**.** *On a compact oriented four-manifold without boundary, a globally exact self-dual two-form vanishes.*

Proof. If $\alpha = d\beta$ and $*\alpha = \alpha$, then $\int |\alpha|^2 dV = \int \alpha \wedge \alpha = \int d(\beta \wedge \alpha) = 0$. □

Our forms are exact on $\mathbb{C}^2$. A nontrivial compact realization must therefore change their global period, boundary or singularity data; a descended form may have a primitive that fails to descend. On the original noncompact space, the forms are independent of w and are not square integrable unless zero. Compact integration by parts cannot be imported without controlling the missing boundaries.

There is a further restriction on extending the entire input family to a compact hyperkähler manifold. With the orientation in which the parallel triple is self-dual, the Weitzenböck formula for a harmonic self-dual form is

$$0 = \nabla^*\nabla\alpha + (\mathrm{Scal}/3 - 2W^+)\alpha = \nabla^*\nabla\alpha.$$

Here scalar curvature and W^+ vanish. Integration implies parallelness. This makes metric-relative pairings constant, although coordinate coefficients of a parallel form need not be constant. For the actual family a stronger local argument applies: at a real point with $H \neq 0$, the forms α_H and α_{H_z} are proportional with a real ratio c. If both were parallel for the same connection, their difference $\alpha_{H_z} - c\alpha_H$ would vanish throughout the connected patch. This forces $H_z = cH$, incompatible with a nonzero even entire field decaying along the real axis. This obstruction concerns preservation of the whole particular family, not every possible compact geometric representation.

5 A holomorphic surface with the physical metric coefficient

Fix real t, abbreviate $f(z) = H(t, z)$, and choose a holomorphic primitive $P' = f$. For the entire theta field a global primitive exists. Define

$$\mathcal{E}(z) = |f(z)|^2 + |f'(z)|^2, \qquad \Omega_t = \{z : \mathcal{E}(z) > 0\}.$$

On $\Omega_t \times \mathbb{C}$ consider

$$X_t(z, w) = (P(z), f(z), w) \in \mathbb{C}^3. \tag{5.1}$$

The complex tangent vectors $(f, f', 0)$ and $(0, 0, 1)$ are independent exactly where $\mathcal{E} > 0$. Thus (5.1) is a holomorphic immersion with induced real metric and Kähler form

$$g_t = \mathcal{E}(z)|dz|^2 + |dw|^2, \tag{5.2}$$

$$\omega_t = \frac{i}{2}\{\mathcal{E}\, dz \wedge d\bar{z} + dw \wedge d\bar{w}\}. \tag{5.3}$$

The parameter domain with this metric is a smooth Riemannian four-manifold. Its image is an immersed complex surface; global injectivity and absence of self-intersections are not asserted. On the physical real slice, $\mathcal{E}(x) = E(t, x)$ exactly. At a common zero of H and H_z, the map loses one complex rank and (5.2) is not positive definite.

5.1 Curvature and induced holonomy

Proposition 5.1 (Curvature of the jet surface)**.** *Set $D = ff'' - (f')^2$. With the real metric convention $|dz|^2 = dx^2 + dy^2$, the Gaussian curvature of the first factor of (5.2) is*

$$K_t = -\frac{2|D|^2}{\mathcal{E}^3}. \tag{5.4}$$

The scalar curvature of the four-dimensional product is $2K_t$. Its Ricci tensor equals K_t times the curve metric in the first factor and zero in the flat factor; its Ricci form is

$$\rho_t = -i\frac{|D|^2}{\mathcal{E}^2}\, dz \wedge d\bar{z}. \tag{5.5}$$

In particular, on the physical line,

$$K_t(x) = -\frac{2\mathcal{L}_0(t, x)^2}{E(t, x)^3}. \tag{5.6}$$

Proof. The two-by-two Hermitian Gram determinant gives

$$\begin{aligned}\partial_z\partial_{\bar{z}} \log \mathcal{E} &= \frac{\mathcal{E}(|f'|^2 + |f''|^2) - |f'\bar{f} + f''\overline{f'}|^2}{\mathcal{E}^2} \\ &= \frac{|ff'' - (f')^2|^2}{\mathcal{E}^2}.\end{aligned}$$

For $ds^2 = \mathcal{E}(dx^2 + dy^2)$ the Gaussian curvature is $-(2\mathcal{E})^{-1}\Delta \log \mathcal{E}$, and $\Delta = 4\partial_z\partial_{\bar{z}}$. This proves (5.4). Product curvature proves the Ricci tensor and scalar statements. Equivalently the Kähler Ricci form is $-i\partial\bar{\partial} \log \det(g_{a\bar{b}})$; the irrelevant constant in the Hermitian determinant disappears upon differentiation, giving (5.5). On the real line $D = -\mathcal{L}_0$, proving the last identity. □

The metric has a parallel splitting into a curve and a flat complex line, so its holonomy is contained in $U(1) \times \{1\} \subset U(2)$. It is Ricci-flat on a connected patch precisely when $D \equiv 0$ there. That condition forces

$$f(z) = Ae^{bz}, \qquad A \neq 0, \tag{5.7}$$

on that patch. Indeed on a subpatch where $f \neq 0$ one has $(f'/f)' = D/f^2 = 0$, and holomorphic identity continuation proves (5.7). If this equality held on an open patch of a theta slice, the identity theorem would extend it to the whole spectral plane. A nonzero even theta field cannot be such an exponential: evenness would force $b = 0$, whereas $H_{zz}(t, 0) = -\int_0^\infty u^2 e^{tu^2}\Phi(u)\, du < 0$. Thus this induced metric is not a hyperkähler metric for the theta field. This is a statement about (5.2), not a prohibition on other auxiliary metrics or other geometric realizations.

The squaring in (5.6) is decisive. A lower or upper curvature norm does not select the favorable sign of $\mathcal{L}_0$. In addition, the factor E^{-3} makes intrinsic curvature very sensitive to degeneration of the physical metric.

5.2 Genuine calibrated inclusions and their limits

Every locally embedded image patch of (5.1) is calibrated by the ambient Kähler calibration $\omega_0^2/2$ and hence is minimal. The relevant algebraic and variational statements are the complex submanifold calibration and the fundamental theorem of calibrations; see [5, Theorems 3.1 and 2.3]. This conclusion uses a proved holomorphic immersion, rather than an analogy between quaternionic coefficients and tangent vectors.

There are literal exceptional-calibration realizations as well. On $\mathbb{R}_r \times \mathbb{C}^3$ use the standard G_2 form

$$\varphi_0 = dr \wedge \omega_0 + \operatorname{Re}\Omega_3, \qquad *_0\varphi_0 = \tfrac{1}{2}\omega_0^2 - dr \wedge \operatorname{Im}\Omega_3.$$

The image of (5.1) in $\{0\} \times \mathbb{C}^3$ is coassociative: the second term vanishes and the first restricts to its induced volume. Alternatively, regard $(P, f, w, 0)$ as a surface in $\mathbb{C}^4 \cong \mathbb{R}^8$. For the standard Cayley calibration

$$\Psi_0 = \tfrac{1}{2}\omega_0^2 + \operatorname{Re}\Omega_4, \tag{5.8}$$

the holomorphic four-form pulls back to zero, while the first term is the volume. Hence this image is Cayley. These standard inclusions are also recorded in [5, Propositions 6.5 and 7.3]. The ambient metrics used here are flat, with trivial holonomy contained in the indicated groups. We have not produced an ambient metric whose holonomy is exactly G_2 or Spin(7).

Calibration does not remove a branch degeneration. If at a fixed time

$$f(z) = c(z-a)^m + O((z-a)^{m+1}), \qquad c \neq 0, \quad m \geq 2,$$

then

$$\mathcal{E}(z) \sim m^2|c|^2|z-a|^{2m-2}. \tag{5.9}$$

The radial distance to a is asymptotic to $|c||z-a|^m$, and the circumference of a small coordinate circle is asymptotic to $2\pi m|c||z-a|^m$. Thus the completion has asymptotic cone angle $2\pi m$ in the curve factor. It is at finite distance but is not a nondegenerate smooth point of the original metric. One can instead make the everywhere immersive graph $(z, P, H, w) \in \mathbb{C}^4$. Its induced metric is $(1+\mathcal{E})|dz|^2 + |dw|^2$, and it remains a calibrated complex surface through a physical joint zero. This repairs immersion by introducing a background metric term; it does not prove nonvanishing of E.

6 The marked real Gauss map retains the sign

The nonconstant tangent direction of (5.1) is the projective Gauss map

$$G_0(t,z) = [H(t,z) : H_z(t,z)] \in \mathbb{CP}^1. \tag{6.1}$$

The second tangent direction is fixed, so this projective line captures the varying part of the full tangent plane. Normalize the Fubini–Study metric by $ds^2_{\mathrm{FS}} = |d\zeta|^2/(1+|\zeta|^2)^2$. Differentiating in either projective chart gives the coordinate-independent identities

$$G_0^* ds^2_{\mathrm{FS}} = \frac{|D|^2}{\mathcal{E}^2}|dz|^2, \qquad \rho_t = -2G_0^*\omega_{\mathrm{FS}}. \tag{6.2}$$

Real reflection puts the physical image in $\mathbb{RP}^1$. Its orientation along the marked real spectral line is extra information which the nonnegative metric in (6.2) omits.

For both branches put

$$E_j^{\mathrm{jet}} = h_j^2 + h_{j+1}^2, \qquad G_j = [h_j : h_{j+1}], \qquad j = 0, 1. \tag{6.3}$$

On a real interval on which this pair never vanishes, choose a continuous real lift $\alpha_j = \arg(h_j + i h_{j+1})$. The lift is defined up to a constant multiple of 2π; its projective value is defined modulo π. Its derivative and every continuously lifted increment are unambiguous. Differentiating the argument proves

$$\boxed{\partial_x \alpha_j = \frac{h_j h_{j+2} - h_{j+1}^2}{h_j^2 + h_{j+1}^2} = -\frac{\mathcal{L}_j}{E_j^{\mathrm{jet}}}.} \tag{6.4}$$

For $j = 0$, $E_0^{\mathrm{jet}} = E$. For $j = 1$ the denominator is different and can vanish even when $E > 0$. No global regularity of G_1 as a physical unit vector is being presumed.

Proposition 6.1 (An oriented angular certificate)**.** *Suppose prescribed functions $\kappa_j \geq 0$ are locally bounded through the covered candidate events and*

$$\partial_x \alpha_j \leq \kappa_j \quad \text{on } S_j \cap \{h_j \neq 0\}, \qquad j = 0, 1.$$

Then $\mathcal{L}_j \geq -\kappa_j E$ on the full stationary sets. Consequently the hypotheses of Theorem 2.5 are satisfied on the same covered domain.

Proof. On S_j with $h_j \neq 0$, (6.4) gives $\mathcal{L}_j \geq -\kappa_j h_j^2 \geq -\kappa_j E$. The second inequality uses both $h_j^2 \leq E$ for $j = 0, 1$ and $\kappa_j \geq 0$. At a remaining point of S_j one has $h_j = 0$, so $\mathcal{L}_j = 0$ and the same inequality holds without defining an angle there. □

For $j = 0$ this certificate is the original signed stationary bound. For $j = 1$ it is in general a stronger sufficient bound, since it pays by h_1^2 on S_1 instead of E. On an odd collision predecessor their ratio tends to one by (2.11). These distinctions are necessary when comparing an auxiliary projective metric with the physical two-component energy.

6.1 A sign-reversing geometric control

Fix $\varepsilon > 0$ and let

$$f_+(z) = \varepsilon + z^2, \quad f_-(z) = \varepsilon - z^2, \qquad P_\pm(z) = \varepsilon z \pm z^3/3.$$

At the real center,

$$E_\pm = \varepsilon^2, \quad \mathcal{L}_{0,+} = -2\varepsilon, \quad \mathcal{L}_{0,-} = 2\varepsilon, \quad K_+ = K_- = -8/\varepsilon^4. \tag{6.5}$$

The corresponding complex surfaces are unitarily congruent after rotating the parameter:

$$\mathrm{diag}(-i, 1, 1)\, X_+(iz, w) = X_-(z, w). \tag{6.6}$$

This follows by substituting iz into each of the two polynomial components. Their intrinsic and calibrated geometry therefore cannot distinguish the two signs without retaining the marked real slice and its orientation. Each polynomial has an exact backward-heat extension $f_\pm(z) \mp 2(t - t_0)$ from its displayed time slice. In particular, $H(t, z) = z^2 - 2(t - T)$ is calibrated at every regular point of its associated surface but develops a double real zero at $(T, 0)$. Its predecessor at $x = 0$ has $\alpha_0' = 1/(T - t)$.

7 Conditional exclusion by signed chords and spatial turning

The marked angle in (6.4) retains a sign that curvature magnitude discards. Its most useful finite-interval observation is a signed chord paid by the physical energy at the stationary starting point. We first prove a multiplicity-independent collision obstruction for that observation, then give the finer turning profiles and the supporting continuity criterion. These are local heat-flow results; the complete-theta estimate required to apply them remains open. The detailed derivations and finite controls are recorded in [9, 10].

7.1 A fixed nearby endpoint and starting-point energy

Put $W = h_j$, $j \in \{0, 1\}$. When $W_x(t, x) = 0$ and W is nonzero on the whole interval $[x, y]$, $y > x$, a continuous angle lift satisfies

$$\alpha_j(t, y) - \alpha_j(t, x) = \arctan \frac{W_x(t, y)}{W(t, y)}. \tag{7.1}$$

Indeed, W has one sign on the interval, so its lifted argument stays in one open half-plane, with no winding ambiguity. In particular,

$$\begin{aligned} \mathcal{C}_j(t; x, y) &:= h_j(t, x)\{h_j(t, y) - h_{j+1}(t, y)\}, \\ \mathcal{C}_j(t; x, y) \geq 0 \quad &\Longleftrightarrow \quad \alpha_j(t, y) - \alpha_j(t, x) \leq \frac{\pi}{4}. \end{aligned} \tag{7.2}$$

All chord losses below use $e = E(t, x)$ at the starting point.

Theorem 7.1 (The fixed-endpoint collision law)**.** *Suppose H has exact spatial multiplicity $2r + j$ at (T, a), $r \geq 1$, and let $W = h_j$, $\chi(s)$ be its full stationary predecessor, and $k = \partial_x^{2r} W(T, a)/r! \neq 0$. For every sufficiently small fixed $\delta > 0$, the entire interval $[\chi(s), a + \delta]$ is W-zero-free for sufficiently small $s > 0$, and*

$$\boxed{\frac{\mathcal{C}_j(T - s; \chi(s), a + \delta)}{\sqrt{E(T - s, \chi(s))}} \longrightarrow -d_\delta < 0,} \tag{7.3}$$
$$d_\delta := -\operatorname{sgn}(k)\{W(T, a + \delta) - W_x(T, a + \delta)\} > 0.$$

Proof. Write $a_0 = \partial_x^{2r} W(T, a)/(2r)!$. The heat Taylor expansion (2.9), applied to the analytic function $W(T - \eta^2, a + z)$, gives the ordinary two-variable remainder

$$W(T - \eta^2, a + z) = a_0 P_{2r}(\eta^2, z) + O((|\eta| + |z|)^{2r+1}), \tag{7.4}$$
$$P_{2r}(\eta^2, z) = \sum_{p=0}^{r} \frac{(2r)!}{p!(2r - 2p)!} \eta^{2p} z^{2r-2p},$$
$$P_{2r}(\eta^2, z) \geq (\eta^2 + z^2)^r \geq 2^{-r}(|\eta| + |z|)^{2r}. \tag{7.5}$$

For the first inequality, the ratio of the coefficient indexed by p to the corresponding binomial coefficient is $\prod_{i=0}^{p-1} 2(2r - 2i - 1) \geq 1$, with empty product one. Thus the remainder divided by P_{2r} is $O(|\eta| + |z|)$. In a sufficiently small fixed neighborhood, W has the sign of a_0 whenever $(\eta, z) \neq (0, 0)$. Since $\chi(s) - a = O(s)$, this proves zero-freeness on the whole asserted interval, not just at its endpoints.

At time T,

$$W(T, a + \delta) - W_x(T, a + \delta) = -2ra_0\delta^{2r-1} + O(\delta^{2r}),$$

which has the opposite sign from k for every sufficiently small positive δ. Finally, (2.8) and (2.11) give

$$W(T-s,\chi(s)) = ks^r(1+O(s)), \qquad E(T-s,\chi(s)) = k^2 s^{2r}(1+O(s)). \tag{7.6}$$

Consequently $W(T-s,\chi(s))/\sqrt{E(T-s,\chi(s))} \to \operatorname{sgn}(k)$; analyticity at the fixed endpoint proves (7.3). □

Theorem 7.2 (Starting-energy chord exclusion)**.** *Let H be a real analytic backward-heat field with no identically zero spatial slice. Near each covered candidate event suppose there are a fixed length $\ell_0 > 0$, $e_0 > 0$, and prescribed nonnegative functions $\omega_j(e) = o(\sqrt{e})$ as $e \downarrow 0$, such that*

$$\mathcal{C}_j(t;x,y) \geq -\omega_j(E(t,x)), \qquad j = 0,1, \tag{7.7}$$

uniformly whenever $h_{j+1}(t,x) = 0$, $0 < y - x \leq \ell_0$, $0 < E(t,x) \leq e_0$, and h_j is nonzero on the whole segment, which must lie in the covered neighborhood. Assume that neighborhood contains the sufficiently close predecessors and these short intervals. Then no covered finite multiple real zero occurs. In particular, a bound $\mathcal{C}_j \geq -K_j E(t,x)^\gamma$ suffices for any fixed $\gamma > 1/2$, with $K_j \geq 0$ locally bounded uniformly in both endpoints. For the complete theta field, coverage of $0 < T \leq 1/2$ implies RH by Remark 2.2.

Proof. At a putative event use its parity branch and choose $\delta > 0$ in Theorem 7.1, smaller than $\ell_0/2$ and inside the covered neighborhood. The lengths $a + \delta - \chi(s)$ are positive and smaller than ℓ_0 for small $s > 0$. Divide (7.7) by the square root of the starting energy and use (7.6). Its lower bound tends to zero, contradicting the strictly negative limit (7.3). At a terminal time a left-hand time neighborhood suffices. □

The fixed positive allowable length and uniformity through the event are essential hypotheses. A bound available only on one prescribed shrinking family need not cover the fixed nearby endpoint used here. Both branches are required because the event multiplicity can have either parity.

Remark 7.3 (Comparison with stationary and angular estimates). At multiplicity $2r + j$, the stationary law and the fixed-endpoint law are

$$\mathcal{L}_j \sim -\text{const}\, E^{1-1/(2r)}, \qquad \mathcal{C}_j \sim -\text{const}\, E^{1/2}.$$

Thus any fixed exponent $\gamma > 1/2$ works in the chord target at every finite multiplicity. For a stationary loss KE^γ, any fixed $\gamma < 1$ eventually becomes too weak as r grows. This comparison does not order the full hypotheses: chord estimates concern whole intervals, whereas Theorem 2.5 imposes a condition only on the stationary sets.

Proposition 7.4 (Sharpness of the energy scale and location)**.** *The exponent $1/2$ with an arbitrary bounded coefficient, or a bounded loss paid by endpoint energy, does not exclude collisions in general.*

Proof. The backward-heat field $H(T-s,x) = x^2 + 2s$ has a double collision, stationary predecessor $x = 0$, and zero-free rightward intervals for every $s > 0$. With $y = \ell \geq 0$,

$$e = E(T-s,0) = 4s^2, \qquad \mathcal{C}_0 = 2s(2s + \ell^2 - 2\ell) \geq -2s = -\sqrt{e}.$$

For endpoint energy put $q = H_x(T-s,\ell)/H(T-s,\ell) \geq 0$ and $b = H(T-s,0)/H(T-s,\ell) \in (0,1]$. Then

$$\frac{\mathcal{C}_0}{E(T-s,\ell)} = b\frac{1-q}{1+q^2} \geq -\frac{1}{4}, \qquad 1 - q + \frac{1+q^2}{4} = \frac{(q-2)^2+1}{4} > 0.$$

For $q \leq 1$ the quotient is nonnegative; for $q > 1$, multiplication by $b \leq 1$ improves the lower bound. The still weaker payment by the sum of starting and endpoint energies also permits this collision. □

7.2 Transfer from a complete family of cutoff estimates

The chord criterion permits quantitative losses that need not be linear in E. The cutoff must be chosen only after a single prescribed family of estimates has been proved for every integer cutoff.

Proposition 7.5 (All-cutoff transfer)**.** *Let Q be a stationary observation with $e = E(t,x)$, or a chord observation with $e = E(t,x)$ at its starting point. Suppose*

$$Q \geq -Ae\exp(aN^{\beta}) - B\exp(-cN^2) \quad \text{for every integer } N \geq N_0, \tag{7.8}$$

where $A, a, B \geq 0$, $c > 0$, and $N_0 \geq 0$ is an integer. Assume fixed local upper bounds A_*, a_*, B_*, N_*, with $N_* \geq 1$, and a positive lower bound c_0 for c. For chords all these bounds are uniform over the admissible intervals of Theorem 7.2. The coefficient rules are prescribed before selecting the cutoff and are not fitted to an observed energy floor.

If $0 < \beta < 2$, set $M = \max(1, B_*)$, $L(e) = 1 + \max(0, \log(M/e))$, and $D = N_* + 1 + c_0^{-1/2}$. For $e > 0$,

$$Q \geq -(A_* + 1)e\exp\{a_* D^{\beta} L(e)^{\beta/2}\}. \tag{7.9}$$

This excludes finite collisions for the stationary target with its predecessor coverage, and for the chord target with its full interval coverage.

If $\beta = 2$ and $a_* < c_0$, put

$$\kappa = a_* + c_0, \qquad p = \frac{c_0}{\kappa} > \frac{1}{2}, \qquad D_2 = N_* + 1.$$

For $0 < e \leq M$, with $L_2 = \log(M/e)$, one has

$$Q \geq -M^{1-p}e^{p}\left[A_*\exp\left\{a_* D_2^2 + 2a_* D_2\sqrt{L_2/\kappa}\right\} + 1\right]. \tag{7.10}$$

This loss is $o(\sqrt{e})$ and therefore excludes collisions for the chord target. Equivalently, every $1/2 < \gamma < p$ yields a bound $Q \geq -K_\gamma e^{\gamma}$ for sufficiently small e.

Proof. For the first claim choose

$$N = \left\lceil \max\left(N_*, \sqrt{\max(0, \log(M/e))/c_0}\right)\right\rceil.$$

Then $B\exp(-cN^2) \leq e$ and $N \leq D\sqrt{L(e)}$, proving (7.9). Its relative loss is $\exp(o(\log(1/e)))$. Along (7.6) it is $s^{-o(1)}$, which cannot absorb the stationary pole r/s. Its absolute loss is also $o(\sqrt{e})$, so Theorem 7.2 applies.

For the Gaussian claim choose $N = \lceil \max(N_*, \sqrt{L_2/\kappa})\rceil$. The inequalities $\sqrt{L_2/\kappa} \leq N \leq D_2 + \sqrt{L_2/\kappa}$ give (7.10) directly. The factor $\exp(O(\sqrt{\log(1/e)}))$ pays integer rounding and the cutoff threshold. It is dominated by $e^{-(p-\gamma)}$ for any fixed $\gamma < p$ as $e \downarrow 0$. At $e = 0$, letting $N \to \infty$ in (7.8) gives $Q \geq 0$. □

The strict local rate gap $a_* < c_0$ cannot be replaced here by a nonuniform pointwise gap that closes at the event. Nor may the rounding factor in (7.10) be dropped when claiming an exact $O(e^p)$ loss. At equality of the rates the preceding double-collision control satisfies, for every real N,

$$e\exp(N^2) + \exp(-N^2) \geq 2\sqrt{e} = 4s, \qquad \mathcal{L}_0 = -4s, \qquad \mathcal{C}_0 \geq -2s.$$

Thus both observations obey (7.8) with $\beta = 2$ and $A = B = a = c = 1$ despite the collision. These transfer statements are conditional analytic implications, not consequences of checking finitely many cutoffs. A separately truncated observation requires its own complete error estimate; a proved error $Q_0\sqrt{e}\exp(-c_1N^2)$ can be paid by $e/2 + (Q_0^2/2)\exp(-2c_1N^2)$.

7.3 The inner profile and quarter-turn concentration

Lemma 7.6 (The real scaled heat profiles)**.** *Use the notation of Theorem 7.1. Uniformly on each fixed compact interval of ξ,*

$$\begin{aligned} W(T-s,\chi(s)+s\xi) &= ks^r\{1+O(s)\},\\ W_x(T-s,\chi(s)+s\xi) &= ks^r\{r\xi+O(s)\},\\ W_{xx}(T-s,\chi(s)+s\xi) &= ks^{r-1}\{r+O(s)\}. \end{aligned} \tag{7.11}$$

On each fixed interval $0\le y\le c$, $c>0$, one also has

$$W(T-s,\chi(s)+\sqrt{s}\,y) = ks^r\{F_r(y)+O(\sqrt{s})\}, \tag{7.12}$$

$$W_x(T-s,\chi(s)+\sqrt{s}\,y) = ks^{r-1/2}\{F_r'(y)+O(\sqrt{s})\}, \tag{7.13}$$

$$F_r(y) = \sum_{\ell=0}^{r}\frac{r!}{(r-\ell)!(2\ell)!}y^{2\ell}. \tag{7.14}$$

In particular, W has the sign of k on each such whole interval for small $s>0$; $F_r\ge 1$ and $F_r'(c)>0$.

Proof. In (2.9) put $x-a=s(\eta(s)+\xi)$, where $\eta(s)=(\chi(s)-a)/s$ is analytic. The leading coefficients of W, W_x, W_{xx} after removing respectively s^r, s^r, s^{r-1} are $k, kr\xi, kr$; for W_x, the constant term cancels because $\eta(0)=-\partial_x^{2r+1}W(T,a)/(r\partial_x^{2r}W(T,a))$. The resulting analytic expansions and their fixed-order derivatives are uniform on compact ξ intervals. For the parabolic scale put $s=\eta^2$ instead. The coefficient of η^{2r} in $W(T-\eta^2,\chi(\eta^2)+\eta y)$ is $kF_r(y)$; the displacement $\chi(\eta^2)-a=O(\eta^2)$ does not affect it. Uniform analytic remainders and their first y derivatives prove the two expansions. Positivity of the displayed polynomial proves the sign assertions. □

Equations (6.4) and (7.11) give the finer angular core, uniformly on compact ξ intervals:

$$\begin{aligned} \alpha_j(T-s,\chi(s)+s\xi)-\alpha_j(T-s,\chi(s)) &= \arctan(r\xi)+O(s),\\ s\,\partial_x\alpha_j(T-s,\chi(s)+s\xi) &= \frac{r}{1+r^2\xi^2}+O(s). \end{aligned} \tag{7.15}$$

These compact-window limits assert no weak limit on a fixed spatial neighborhood and require no unproved uniform tail estimate. For an odd collision the same Taylor counting gives $H(T-s,\chi(s)+s\xi)=O(s^{r+1})$, so

$$\frac{E_1^{\mathrm{jet}}}{E}(T-s,\chi(s)+s\xi)\longrightarrow 1+r^2\xi^2. \tag{7.16}$$

Thus agreement of the two energies at the stationary center cannot be extended across a nonzero inner window without retaining this factor.

Theorem 7.7 (Positive quarter-turn concentration)**.** *For each fixed $c>0$, the collision and predecessor above satisfy*

$$\alpha_j(T-s,\chi(s)+c\sqrt{s})-\alpha_j(T-s,\chi(s))\longrightarrow\frac{\pi}{2}. \tag{7.17}$$

The defining pair is regular on the whole interval, including on branch $j=1$. The chord limits at the two shrinking scales are

$$\begin{aligned} \frac{\mathcal{C}_j(T-s;\chi(s),\chi(s)+Cs/r)}{E(T-s,\chi(s))} &\longrightarrow 1-C\quad(C>0),\\ \frac{\mathcal{C}_j(T-s;\chi(s),\chi(s)+c\sqrt{s})}{E(T-s,\chi(s))} &= -\frac{F_r'(c)}{\sqrt{s}}+O(1). \end{aligned} \tag{7.18}$$

Proof. Zero-freeness follows from Lemma 7.6. At the parabolic endpoint,

$$W_x/W = s^{-1/2}(F_r'(c)/F_r(c))(1 + O(\sqrt{s})) \to +\infty.$$

Equation (7.1) proves the angle limit with either sign of k. Substitution of the two jet expansions into (7.2), using (7.6), proves both chord limits. □

The finite inner limit alone can be absorbed by a growing relative energy allowance. A bound uniform on all short intervals must also hold on the parabolic windows and at the fixed nearby endpoint, where the stronger obstructions apply. This interval quantifier explains why the cutoff conclusions above are stronger than a test on one fixed inner window; see [10].

7.4 A supporting one-sided continuity criterion

Definition 7.8 (One-sided modulus from stationary starting points)**.** A field has a local one-sided turning modulus on a declared domain if, near each covered candidate event, there are a fixed $\ell_0 > 0$ and a prescribed nonnegative function $\omega(\ell) \to 0$ as $\ell \downarrow 0$, uniform in time and starting point, such that for $j = 0, 1$

$$\alpha_j(t, x + \ell) - \alpha_j(t, x) \le \omega(\ell) \tag{7.19}$$

whenever $0 < \ell \le \ell_0$, $h_{j+1}(t, x) = 0$, $h_j(t, x) \neq 0$, the segment lies in the covered neighborhood, and $(h_j, h_{j+1}) \neq (0, 0)$ throughout it. The increment uses its continuous lift along the segment.

Theorem 7.9 (Oriented-continuity exclusion)**.** *Let H be a real analytic backward-heat field with nonzero spatial slices. If Definition 7.8 holds with coverage of the sufficiently close full predecessors and their short rightward intervals, then no covered finite multiple real zero occurs. For the complete theta field, coverage of $0 < T \le 1/2$ implies RH by Remark 2.2.*

Proof. At a candidate event take $\ell = c\sqrt{s}$ on its parity branch. Lemma 7.6 supplies the required regularity on the whole interval. The upper bound tends to zero, contradicting Theorem 7.7. □

On every regular test interval the observed increment is exactly

$$\int_x^{x+\ell} \frac{h_j h_{j+2} - h_{j+1}^2}{h_j^2 + h_{j+1}^2}(t, y)\, dy. \tag{7.20}$$

A uniform upper bound on the angular derivative throughout these intervals implies a linear modulus. More generally a uniform strict gap below $\pi/2$ on all sufficiently short regular intervals already contradicts (7.17). Both formulations retain the orientation and allow negative turning; neither follows from a curvature-magnitude bound or from continuity on individual regular slices. No global regularity of the second branch is assumed.

The resulting hierarchy identifies concrete conditional source targets, with (7.7) the most tolerant energy-scale target proved here. The exact polynomial controls in [9, 10] test the local identities and their scope; they establish no sign bound for the theta source. The remaining task is a complete-theta estimate with the required signs, starting-point energy, and uniform interval coverage.

8 The complete theta source on a curved elliptic family

The preceding geometric realizations retain the physical jets, but a local realization alone does not distinguish the theta source from a collision control. We now place the complete source in a family carrying its actual integer lattice. There are three separate structures: the holomorphic theta

line bundle, the physical heat-weighted transform, and a nonflat hyperkähler metric on the elliptic family. Their relations below are exact. They do not yet give the signed estimate at the physical energy scale. Our source normalization is that of [1].

8.1 The complete primitive and its boundary contribution

For $u \geq 0$ set

$$\Phi(u) = \sum_{n\geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \qquad H(t,x) = \int_0^\infty e^{tu^2} \Phi(u) \cos(xu)\, du. \tag{8.1}$$

Every fixed source derivative decays faster than any Gaussian as $u \to +\infty$. The sums and integrals below therefore converge locally uniformly with all stated derivatives. In particular, H is entire in $(t,x) \in \mathbb{C}^2$ and satisfies $H_t = -H_{xx}$.

Write

$$\vartheta(z,\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2 \tau + 2\pi i n z}, \qquad \Im\tau > 0, \quad P(u) = e^u \vartheta(0, ie^{4u}), \quad \psi(u) = e^u \sum_{n\geq 1} e^{-\pi n^2 e^{4u}}. \tag{8.2}$$

Thus $P = e^u + 2\psi$ on the half-line. The Gaussian Poisson identity $\vartheta(0, i/y) = \sqrt{y}\,\vartheta(0, iy)$ implies $P(-u) = P(u)$. The function P grows like e^u at positive infinity; it is ψ, not P, that supplies a convergent positive-time primitive transform.

Proposition 8.1 (Complete differential reconstruction)**.** *The complete source and primitive satisfy*

$$\Phi = \frac{P'' - P}{16} = \frac{\psi'' - \psi}{8}, \qquad \psi^{(2k+1)}(0) = -\frac{1}{2} \quad (k \geq 0). \tag{8.3}$$

For

$$J(t,x) = \int_0^\infty e^{tu^2} \psi(u) \cos(xu)\, du \tag{8.4}$$

one has the exact identity

$$\boxed{8H = \frac{1}{2} + (2t - x^2 - 1)J + 4txJ_x - 4t^2 J_{xx}.} \tag{8.5}$$

All the theta indices occur in these identities.

Proof. Termwise differentiation of P gives the first identity in (8.3); $(e^u)'' - e^u = 0$ gives the second expression. Evenness of $P = e^u + 2\psi$ gives every displayed endpoint derivative. For $F(u) = e^{tu^2}\cos(xu)$, two integrations by parts yield

$$\int_0^\infty \psi'' F\, du = -\psi'(0)F(0) + \psi(0)F'(0) + \int_0^\infty \psi F''\, du.$$

The upper boundary terms vanish, $F(0) = 1$, and $F'(0) = 0$. Directly,

$$F'' - F = e^{tu^2}\big[(2t + 4t^2u^2 - x^2 - 1)\cos(xu) - 4txu\sin(xu)\big].$$

Insert $\psi'(0) = -1/2$ and differentiate (8.4). This gives (8.5), including its boundary constant. □

Proposition 8.2 (Positive primitive; signed reconstruction)**.** *For $0 \le t \le 1/2$ and every real x, $J(t,x) > 0$. Uniformly on this heat interval,*

$$J(t,x) = \frac{1}{2x^2} - \frac{1+6t}{2x^4} + O(x^{-6}) \qquad (|x| \to \infty). \tag{8.6}$$

At time zero the exact reconstruction is

$$J(0,x) = \frac{1}{2(1+x^2)} - \frac{8H(0,x)}{1+x^2}. \tag{8.7}$$

Proof. Let $q_n = \pi n^2 e^{4u}$ and $f_n(u) = \exp(tu^2 + u - q_n)$, so that $f = e^{tu^2}\psi = \sum_{n\ge1} f_n$. For $u \ge 0$ and $0 \le t \le 1/2$, $2tu + 1 \le u + 1 \le q_n$. Hence

$$\frac{f_n''}{f_n} = (2tu + 1 - 4q_n)^2 + 2t - 16q_n \ge 9q_n^2 - 16q_n > 0.$$

Thus $f'' > 0$. Moreover $f'(0) = -1/2$ and $f'(\infty) = 0$. For $x \ne 0$, integration by parts gives

$$J(t,x) = \frac{1}{x^2}\int_0^\infty f''(u)(1 - \cos(xu))\,du > 0. \tag{8.8}$$

At $x = 0$ the defining integral is positive. Repeated integration by parts uses $f'(0) = -1/2$ and $f'''(0) = -(1+6t)/2$ to give (8.6). The sixth derivative is uniformly integrable on the stated compact heat interval, which bounds its remainder. Finally (8.7) is (8.5) with $t = 0$. □

Remark 8.3 (What this positivity does not prove). The physical field is reconstructed from a signed differential expression in a strictly positive channel. At $t = 0$ it is the difference between that channel and an explicit rational profile. The primitive-compatible collision controls of [3], have the same boundary constant, strict primitive positivity, and every algebraic asymptotic coefficient of J, while possessing a positive-time double zero. Their source is not the integer theta series. Accordingly, the next construction retains the theta line bundle and its section, rather than using primitive positivity alone.

8.2 The theta line bundle, its connection, and the fourth jet

Use the marked elliptic family

$$\mathcal{X} = (\mathbb{H} \times \mathbb{C})/\mathbb{Z}^2, \qquad (m,n) : (\tau, z) \longmapsto (\tau, z + m + n\tau),$$

where $\mathbb{H}$ is the upper half-plane. Write $\tau = p + iq$, $z = r + \tau s$, so that $q > 0$ and r, s are real fiber coordinates modulo one. The factors

$$j_{m,n}(\tau, z) = e^{-\pi i n^2 \tau - 2\pi i n z} \tag{8.9}$$

define a holomorphic line bundle $L \to \mathcal{X}$, and reindexing the series gives

$$\vartheta(z + m + n\tau, \tau) = j_{m,n}(\tau, z)\vartheta(z, \tau), \qquad 4\pi i\,\partial_\tau \vartheta = \partial_z^2 \vartheta. \tag{8.10}$$

The second equality follows term by term. These standard theta-section and heat-connection structures are described in [6, Section 6]; the explicit normalizations needed here are derived below.

Ordinary derivatives are jets, not separate copies of the same holomorphic section. For every nonnegative integer k,

$$(\partial_z^k \vartheta)(z+m+n\tau,\tau) = j_{m,n}(\tau,z)\sum_{a=0}^{k}\binom{k}{a}(-2\pi i n)^{k-a}(\partial_z^a\vartheta)(z,\tau). \tag{8.11}$$

Thus a global description must keep the lower-jet terms or choose a connection. The following choice also identifies the actual curvature of the theta line bundle.

Proposition 8.4 (Hermitian theta connection)**.** *In the holomorphic frame on $\mathbb{H}\times\mathbb{C}$, the metric*

$$h(\tau,z) = \exp\left(-\frac{2\pi(\Im z)^2}{q}\right) \tag{8.12}$$

descends to L. Its Chern connection and curvature are

$$\nabla = d + A, \qquad A = \partial \log h = 2\pi i s\, dz - \pi i s^2\, d\tau, \tag{8.13}$$

$$F_\nabla = \frac{\pi}{q}\eta\wedge\overline{\eta} = -2\pi i\, dr\wedge ds, \qquad \eta = dz - s\, d\tau = dr + \tau ds. \tag{8.14}$$

In particular, $\int_{E_\tau} iF_\nabla/(2\pi) = 1$. The vertical covariant operator in the chosen relative frame is

$$\mathcal{D} = \nabla_{\partial_z} = \partial_z + 2\pi i s. \tag{8.15}$$

Each $\mathcal{D}^k\vartheta$ transforms by the multiplier (8.9).

Proof. Put $v = \Im z$. Under a lattice translation v becomes $v + nq$, while $|j_{m,n}|^2 = e^{2\pi n^2 q + 4\pi n v}$. Consequently $h(\tau, z+m+n\tau)|j_{m,n}(\tau,z)|^2 = h(\tau,z)$. Using $\partial_z v = 1/(2i)$ and $\partial_\tau q = 1/(2i)$ gives (8.13). Differentiation yields

$$dA = 2\pi i\, ds\wedge(dz - s\, d\tau).$$

Since $ds = (\eta - \overline{\eta})/(2iq)$, this is $(\pi/q)\eta\wedge\overline{\eta}$. Finally $\eta\wedge\overline{\eta} = -2iq\, dr\wedge ds$. The fiber integral is one, with orientation $dr\wedge ds$. Under translation s becomes $s+n$ and $\partial_z j_{m,n} = -2\pi i n j_{m,n}$. The two extra terms cancel in $\mathcal{D}(j_{m,n}a)$ after using the translated s. Iterating proves the covariant transformation statement. □

Remark 8.5. The covariant derivatives here are smooth sections. They need not be holomorphic: $[\partial_{\bar z}, \mathcal{D}] = -\pi/q$. Also, $\mathcal{D}^k$ means repeated vertical differentiation in the translation-invariant frame ∂_z of the relative tangent bundle. It is not an unspecified full covariant Hessian on the total space. The Chern connection is distinct from the theta heat connection in (8.10).

Theorem 8.6 (Exact covariant theta-source formula)**.** *Let $y = e^{4u}$ and evaluate the theta section at $(\tau,z) = (iy,0)$. Then*

$$\Phi(u) = y^{1/4}\left(y^2\partial_y^2 + \frac{3}{2}y\partial_y\right)\vartheta(0,iy) \tag{8.16}$$

$$= y^{1/4}\left(\frac{y^2}{16\pi^2}\partial_z^4 + \frac{3y}{8\pi}\partial_z^2\right)\vartheta(z,iy)\big|_{z=0} \tag{8.17}$$

$$= \boxed{\frac{y^{1/4}}{16\pi^2}(y^2\mathcal{D}^4\vartheta - 3\pi^2\vartheta)(0,iy).} \tag{8.18}$$

The final expression is evaluated in the stated holomorphic frame along the zero section of the marked family. It is a source identity, not a norm identity or a positivity statement.

Proof. For $D_y = y\partial_y$, one has $\partial_u = 4D_y$ and $y^{-1/4} D_y y^{1/4} = D_y + 1/4$. Thus

$$\frac{1}{16} y^{-1/4}(\partial_u^2 - 1) y^{1/4} = D_y^2 + \tfrac{1}{2} D_y = y^2 \partial_y^2 + \tfrac{3}{2} y \partial_y,$$

which proves (8.16). The theta heat equation gives $\partial_y \vartheta(z, iy) = (4\pi)^{-1} \partial_z^2 \vartheta(z, iy)$, proving (8.17). At fixed τ, let $a = 2\pi i s$, so $\mathcal{D} = \partial_z + a$. Then $\partial_z a = \pi/q$ and $\partial_z^2 a = 0$. At $z = 0$, $a = 0$, and the operator product gives

$$(\mathcal{D}^2 \vartheta)|_0 = (\partial_z^2 \vartheta)|_0 + \frac{\pi}{q} \vartheta|_0,$$
$$(\mathcal{D}^4 \vartheta)|_0 = (\partial_z^4 \vartheta)|_0 + \frac{6\pi}{q} (\partial_z^2 \vartheta)|_0 + \frac{3\pi^2}{q^2} \vartheta|_0.$$

Set $q = y$ and substitute the last equality into (8.17). The second-derivative terms cancel, leaving (8.18). □

The physical reconstruction is now an exact integral along the real path $\tau = iy$, $y \geq 1$, in this marked family:

$$H(t, x) = \frac{1}{4} \int_1^\infty e^{t(\log y)^2/16} y^{1/4} \left(y^2 \partial_y^2 + \frac{3}{2} y \partial_y \right) \vartheta(0, iy) \cos\left(\frac{x}{4} \log y \right) \frac{dy}{y} \tag{8.19}$$
$$= \frac{1}{64\pi^2} \int_1^\infty e^{t(\log y)^2/16} y^{1/4} (y^2 \mathcal{D}^4 \vartheta - 3\pi^2 \vartheta)(0, iy) \cos\left(\frac{x}{4} \log y \right) \frac{dy}{y}. \tag{8.20}$$

Both integrals retain all theta indices. The physical heat multiplier $e^{t(\log y)^2/16}$ is not the parameter evolution in the Jacobi heat equation. Inversion $y \mapsto 1/y$ preserves this squared-log multiplier; that reflection symmetry must not be discarded. Nevertheless, the oscillatory integral (8.20) does not become a positive Hermitian norm merely because L has positive fiber degree.

8.3 A nonflat hyperkähler metric with the correct elliptic fibers

The same marked family has an explicit metric which incorporates its varying elliptic modulus. This is a local semi-flat construction in the sense of [7, Section 2]; we give a direct verification so that neither its curvature nor its global scope is implicit.

Proposition 8.7 (Semi-flat geometry on the marked cover)**.** *On $\mathbb{H} \times (\mathbb{R}/\mathbb{Z})^2$ the metric*

$$g = q(dp^2 + dq^2) + q^{-1}(dr + p\, ds)^2 + q\, ds^2 = q|d\tau|^2 + q^{-1}|\eta|^2 \tag{8.21}$$

is hyperkähler and nonflat. Its elliptic fibers have modulus τ and area one. A parallel hyperkähler triple is

$$\omega_I = q\, dp \wedge dq + dr \wedge ds, \qquad \omega_J + i\omega_K = d\tau \wedge dz. \tag{8.22}$$

For the orientation $\omega_I^2/2 > 0$, the three forms are self-dual. The horizontal sectional curvature is $1/(2q^3)$.

Proof. The fiber metric is $q^{-1}|dr + \tau ds|^2$, with area form $dr \wedge ds$. Introduce the orthonormal coframe

$$e^1 = \sqrt{q}\, dp, \quad e^2 = \sqrt{q}\, dq, \quad e^3 = q^{-1/2}(dr + p\, ds), \quad e^4 = \sqrt{q}\, ds.$$

Writing $e^{ab} = e^a \wedge e^b$, direct expansion gives

$$\omega_I = e^{12} + e^{34}, \qquad \omega_J = e^{13} - e^{24}, \qquad \omega_K = e^{14} + e^{23}, \qquad \omega_a \wedge \omega_b = 2\delta_{ab} \operatorname{vol}_g . \tag{8.23}$$

The first form is closed. Moreover $d\tau \wedge dz = d\tau \wedge (dr + \tau ds)$, which is closed, so the other two forms are closed as well.

For a direct Kähler verification, (τ, z) are holomorphic coordinates on a local lift, and

$$\omega_I = \frac{i}{2}(q\, d\tau \wedge d\bar{\tau} + q^{-1}\eta \wedge \bar{\eta}).$$

The Hermitian coefficient matrix of the metric is

$$G = \begin{pmatrix} q + s^2/q & -s/q \\ -s/q & 1/q \end{pmatrix}, \qquad \det G = 1. \tag{8.24}$$

Thus the Kähler Ricci form $-i\partial\bar{\partial} \log\det G$ vanishes. The induced Chern connection on the canonical bundle has zero connection form in the frame $d\tau \wedge dz$; since the metric is Kähler, this frame is Levi-Civita parallel. Its real and imaginary parts, together with the parallel Kähler form, give the parallel triple (8.22). The algebra (8.23) fixes its normalization and orientation.

Finally, the surfaces with r, s constant are totally geodesic: in these real coordinates the metric is block diagonal, and its base block is independent of r, s. Their induced metric is $q(dp^2 + dq^2)$. Its Gaussian curvature is

$$K_{\text{base}} = -\frac{1}{2q}(\partial_p^2 + \partial_q^2)\log q = \frac{1}{2q^3} > 0. \tag{8.25}$$

Total geodesy identifies this with the ambient sectional curvature of the horizontal plane, proving nonflatness. □

Remark 8.8 (The precise global scope). The formulas descend under the fiber lattice and are defined over the whole marked upper half-plane. They do not give a metric on the full modular quotient. Under

$$\tau' = \frac{a\tau + b}{c\tau + d}, \qquad z' = \frac{z}{c\tau + d}, \qquad ad - bc = 1,$$

one has

$$d\tau' \wedge dz' = (c\tau + d)^{-3} d\tau \wedge dz, \qquad q'|d\tau'|^2 = |c\tau + d|^{-6} q|d\tau|^2. \tag{8.26}$$

The base metric therefore fails modular invariance as written. There is also incompleteness: a path with p, r, s constant and $q \downarrow 0$ has finite length $\int_0^{q_0} \sqrt{q}\, dq$. No complete K3 metric or extension through singular fibers is asserted. The global construction in [7] introduces local models and corrections around singular elliptic fibers; that separate existence theorem does not identify the physical transform with a K3 invariant. Any modular descent here must supply the missing base-period and line-bundle data and then control the resulting boundary terms.

8.4 An actual abelian anti-self-dual theta connection

The connection in Proposition 8.4 carries the arithmetic section, but its curvature is not anti-self-dual for (8.21). This distinction can be resolved explicitly on the marked noncompact family, without changing the vertical source formula.

Proposition 8.9 (A Chern connection with anti-self-dual curvature)**.** *Let $\omega_b = q\, dp \wedge dq$ and $\omega_f = dr \wedge ds$. Equip the same holomorphic theta line bundle with the Hermitian metric*

$$\widehat{h} = \exp\left(\frac{2\pi}{3} q^3\right) h. \tag{8.27}$$

Its Chern connection has connection form and curvature

$$\widehat{A} = A - \pi i q^2\, d\tau, \tag{8.28}$$

$$F_{\widehat{\nabla}} = 2\pi i(\omega_b - \omega_f), \qquad *F_{\widehat{\nabla}} = -F_{\widehat{\nabla}}. \tag{8.29}$$

It is an abelian Yang–Mills connection, and its vertical covariant operator remains $\mathcal{D}$. Thus (8.18) is unchanged.

Proof. The extra factor in (8.27) depends only on the base, so it preserves the lattice descent and the holomorphic bundle. Since $\partial_\tau q = 1/(2i)$,

$$\partial\left(\frac{2\pi}{3}q^3\right) = -\pi i q^2\, d\tau, \qquad d(-\pi i q^2\, d\tau) = 2\pi i q\, dp \wedge dq.$$

Combine this with (8.14). In the orthonormal coframe, $\omega_b = e^{12}$ and $\omega_f = e^{34}$, so $*\omega_b = \omega_f$ and $*\omega_f = \omega_b$. This proves anti-self-duality. The curvature is closed, and hence $d * F_{\widehat{\nabla}} = 0$; for a line bundle this is the Yang–Mills equation. The extra connection form vanishes on ∂_z, proving the last assertion. □

Remark 8.10 (Scope of the ASD construction). This is a nonflat geometric setting with the actual theta section and an anti-self-dual line-bundle connection. The unmodified Chern curvature has self-dual part $F^+_{\nabla} = -\pi i \omega_I$. The base rescaling removes that part while preserving the vertical source expression. It provides neither compactness nor a finite-energy instanton: the real curvature $F_{\widehat{\nabla}}/i$ has squared norm $8\pi^2$, while $\mathrm{vol}_g = q\, dp\, dq\, dr\, ds$ has infinite integral on this cover. The displayed metric's rapid cusp growth is not an invariant obstruction; the classification below gives a representative normalized on the physical path. Neither positive degree nor the Yang–Mills equation supplies the required signed physical estimate.

8.5 Physical restriction and complete signed source identities

The ASD construction does not determine the norm of the distinguished arithmetic section, and its physical restriction is flat. The useful data that survive are the transverse jet identities and the complete signed transform, including its endpoints [11, 12].

Proposition 8.11 (Metric freedom and transverse jets)**.** *Among base-only changes $h_b = e^{b(p,q)}h$, the ASD metrics are exactly*

$$b = \frac{2\pi}{3}q^3 + 2\Re f(\tau), \qquad f \text{ holomorphic on } \mathbb{H}. \tag{8.30}$$

They have the same curvature and vertical operator $\mathcal{D}$. In particular, $f = -\pi i \tau^3/3$ gives

$$\widetilde{h} = e^{2\pi p^2 q}h, \qquad \widetilde{A} = A - \pi i(q^2+\tau^2)d\tau, \qquad \gamma^*\widetilde{h} = 1, \quad \gamma^*\widetilde{A} = 0, \tag{8.31}$$

where $\gamma(u) = (ie^{4u}, 0)$. The natural external product connection on two physical paths is also flat.

In a marked holomorphic chart set $X = 4q\partial_q|_{p,z,\bar{z}}$ and $J_k(u) = (\mathcal{D}^k\vartheta)(0, iq)$, $q = e^{4u}$. Compute the commutators before restriction to the zero section. Then

$$[\nabla^{(b)}_X, \mathcal{D}^k] = -4k\pi i s\mathcal{D}^{k-1} - \frac{k(k-1)\pi}{q}\mathcal{D}^{k-2}, \tag{8.32}$$

$$J_k' = \frac{q}{\pi}J_{k+2} - (2k+1)J_k - \frac{k(k-1)\pi}{q}J_{k-2}. \tag{8.33}$$

Terms with zero prefactor are omitted. The extension X is chart-local; it is not asserted to be a global horizontal field under the τ-dependent lattice translations.

Proof. Direct differentiation gives

$$F_b = -2\pi i\, dr \wedge ds + \frac{i}{2}(b_{pp} + b_{qq})\, dp \wedge dq.$$

Thus ASD is equivalent to $\Delta b = 4\pi q$. The difference from $2\pi q^3/3$ is harmonic, and the simply connected upper half-plane supplies its harmonic conjugate. The stated choice of f gives (8.31). For comparison, $\gamma^* \widehat{A} = d(\pi e^{12u}/3)$; on the external product the connection is the sum of these two exact one-forms, with zero curvature.

Write $a = 2\pi i s$ and $c = \pi/q$. One has $[\nabla_X^{(b)}, \mathcal{D}] = -2a$, $[\mathcal{D}, a] = c$, and $[\mathcal{D}, c] = 0$; the base-only connection term commutes with $\mathcal{D}$. Iteration gives (8.32). The recurrence follows by differentiating the generating identity

$$\sum_{k\geq 0} J_k(u)\frac{w^k}{k!} = e^{\pi w^2/(2q)}\vartheta(w, iq)$$

and using the Jacobi heat equation. The base scalar cancels upon returning to coefficients in the marked arithmetic frame. □

The rapid cusp growth of $\widehat{h}$ is therefore a property of that specified metric, not an obstruction determined by ASD curvature. Changing the metric while keeping ϑ fixed changes its norm; the equivalent holomorphic bundle isometry instead transforms both the section and its dual observation. Flat physical restriction does not remove the second term of (8.32): for $k \geq 2$ it survives at $s = 0$. It also does not make the oscillatory observation a Hermitian norm.

Proposition 8.12 (The complete signed bilocal observations)**.** *Put* $c_j(x, u) = u^j \cos(xu + j\pi/2)$ *and define the chord*

$$\mathcal{C}_j(t; x, y) = h_j(t, x)\{h_j(t, y) - h_{j+1}(t, y)\}, \qquad j = 0, 1.$$

For $x < y$, $h_{j+1}(t, x) = 0$, *and* h_j *nonzero on the whole interval, its nonnegativity is equivalent to an angular increase of at most* $\pi/4$. *Define the symmetric source kernels*

$$K_j(x; u, v) = \frac{(uv)^j}{4}\left[(u+v)^2\cos(x(u-v)) + (-1)^j(u-v)^2\cos(x(u+v))\right], \tag{8.34}$$

$$M(x; u, v) = \frac{1}{2}\left[(1+uv)\cos(x(u-v)) + (1-uv)\cos(x(u+v))\right], \tag{8.35}$$

$$\begin{aligned} T_j(x, y; u, v) = \frac{1}{2}\big[&c_j(x, u)\{c_j(y, v) - c_{j+1}(y, v)\} \\ &+ c_j(x, v)\{c_j(y, u) - c_{j+1}(y, u)\}\big]. \end{aligned} \tag{8.36}$$

With $\mathcal{B}(u) = e^{8u}J_4(u) - 3\pi^2 J_0(u)$, *all three observations have the same complete covariant insertion:*

$$\begin{pmatrix} \mathcal{L}_j(t, x) \\ E(t, x) \\ \mathcal{C}_j(t; x, y) \end{pmatrix} = \frac{1}{256\pi^4}\int_0^\infty\int_0^\infty e^{t(u^2+v^2)+u+v}\mathcal{B}(u)\mathcal{B}(v)\begin{pmatrix} K_j(x; u, v) \\ M(x; u, v) \\ T_j(x, y; u, v) \end{pmatrix} du\, dv. \tag{8.37}$$

Proof. Write $h_j = \int_0^\infty e^{tu^2}\Phi(u)c_j(x, u)\, du$. Multiply the defining observations and symmetrize in u, v. The product-to-sum identities give K_j and M; the directed chord gives T_j. Insert (8.18) in both variables. Complete-source decay justifies every product and differentiation locally uniformly in the parameters. The angular statement follows from the continuous lift $\alpha_j(t, y) - \alpha_j(t, x) = \arctan(h_{j+1}(t, y)/h_j(t, y))$, because h_j has one sign throughout the interval. □

The primitive in (8.2) displays every lower-boundary contribution without splitting the cancelling theta terms. Define

$$I_j(t,x) = \int_0^\infty \psi(u)(\partial_u^2 - 1)\{e^{tu^2} c_j(x,u)\}\, du. \tag{8.38}$$

Two integrations by parts, using $\psi'(0) = -1/2$, give

$$\int_0^\infty (\psi'' - \psi) f\, du = \tfrac{1}{2} f(0) + \psi(0) f'(0) + \int_0^\infty \psi(f'' - f)\, du.$$

Here $f = e^{tu^2} c_j(x,u)$ is even, $f(0) = \delta_{j0}$, and the upper endpoints vanish by double exponential decay. Consequently $8h_j = I_j + \delta_{j0}/2$, and the full signed reconstructions are

$$64\mathcal{L}_0 = I_1^2 - I_0 I_2 - \tfrac{1}{2} I_2, \qquad 64\mathcal{L}_1 = I_2^2 - I_1 I_3, \tag{8.39}$$

$$64E = \tfrac{1}{4} + I_0 + I_0^2 + I_1^2, \tag{8.40}$$

$$\begin{aligned} 64\mathcal{C}_0(t;x,y) &= I_0(t,x)\{I_0(t,y) - I_1(t,y)\} \\ &\quad + \tfrac{1}{2} I_0(t,x) + \tfrac{1}{2}\{I_0(t,y) - I_1(t,y)\} + \tfrac{1}{4}, \end{aligned} \tag{8.41}$$

$$64\mathcal{C}_1(t;x,y) = I_1(t,x)\{I_1(t,y) - I_2(t,y)\}. \tag{8.42}$$

Unspecified arguments in (8.39)–(8.40) are (t,x). The affine terms and constants record the lower edges and corner of the bilocal integration by parts; none are disposable. Every I_j retains the physical heat multiplier inside its differential operator. These identities specify complete quantities to estimate, but assert no sign. In a chord estimate the comparison energy is $E(t,x)$ at the starting point, also on branch $j = 1$; neither endpoint energy nor the auxiliary norm $h_1^2 + h_2^2$ may replace it.

9 Arithmetic discrimination: support, endpoints and reflection

The signed bridge requires information that distinguishes the actual theta section from positive weighted square sums. The following control rejects support-only moment certificates, even with the first source endpoint and primitive normalization imposed. Complete reflection, by contrast, forces the theta coefficients themselves. Neither statement proves the remaining signed estimate.

9.1 A validated collision within the square-support relaxation

Put $\lambda_n = \pi n^2$, $q = e^{4u}$, and

$$\phi_n(u) = e^u(2\lambda_n^2 q^2 - 3\lambda_n q)e^{-\lambda_n q}, \qquad H_n(t,x) = \int_0^\infty e^{tu^2}\phi_n(u)\cos(xu)\, du.$$

For fixed nonnegative weights a_n, write $\Phi_a = \sum_{n\ge1} a_n\phi_n$, $\psi_a = e^u \sum_{n\ge1} a_n e^{-\lambda_n q}$, and $H_a = \sum a_n H_n$. For finite support these definitions give entire physical transforms, $H_{a,t} = -H_{a,xx}$, and $8\Phi_a = \psi_a'' - \psi_a$. The source is positive on $u \ge 0$ whenever some weight is positive. Moreover, for every real polynomial V and every polynomial Q nonnegative on the square nodes,

$$\sum_{n\ge1} a_n e^{-\lambda_n q} Q(\lambda_n) V(\lambda_n)^2 \ge 0 \qquad (q > 0). \tag{9.1}$$

This includes every finite moment matrix and adjacent-square localizing matrix, by taking $Q(\lambda) = (\lambda - \lambda_m)(\lambda - \lambda_{m+1})$.

Proposition 9.1 (Validated weighted-square control)**.** *There are fixed positive weights on $n = 1, 2, 3$, zero otherwise, and a fixed positive normalization such that*

$$\Phi_a'(0) = 0, \qquad \psi_a'(0) = \psi_a'''(0) = -\tfrac{1}{2}, \qquad H_a(\tfrac{1}{4}, x_*) = H_{a,x}(\tfrac{1}{4}, x_*) = 0, \quad H_{a,xx}(\tfrac{1}{4}, x_*) \neq 0,$$

where $44.1303 < x_* < 44.1304$. The third source derivative is nonzero, so this source is not even in u. The control satisfies (9.1) at every degree and every $q > 0$.

Proof. The construction and validated record are given in [14]. Define

$$P_1(a) = -8a^3 + 30a^2 - 15a, \quad d_n = P_1(\lambda_n), \quad G_n = e^{\lambda_n} H_n, \quad b_n = -d_1/d_n \quad (n = 2, 3).$$

Here $d_1 > 0 > d_2, d_3$, and $G_\theta = G_1 + \theta b_2 G_2 + (1-\theta) b_3 G_3$ has positive weights for $0 < \theta < 1$. Its source derivative at zero is exactly $d_1 + \theta b_2 d_2 + (1-\theta) b_3 d_3 = 0$. At $T = 1/4$, set $A = G_1 + b_3 G_3$, $B = b_2 G_2 - b_3 G_3$, and $D = A'B - AB'$. On $I = [44.1303, 44.1304]$, the validated enclosures include

$$A < 0 < B, \qquad D(44.1303) \in (5.0206, 5.0208)10^{-14},$$
$$D' \in (-1.178, -1.121)10^{-9}, \qquad D(44.1304) \in (-6.4752, -6.4750)10^{-14}.$$

Thus D has exactly one zero x_* in I. At that zero, $\theta_* = -A(x_*)/B(x_*) \in (0.87916, 0.87919)$ gives a double zero, with

$$\partial_x^2 G_{\theta_*}(T, x_*) \in (-5.95, -5.65)10^{-6}.$$

The raw weights are $(a_1, a_2, a_3) = (1, \theta_* b_2 e^{3\pi}, (1-\theta_*) b_3 e^{8\pi})$. They are fixed after this construction. Multiplication by $c = -1/(2\psi_a'(0)) > 0$ imposes the stated primitive endpoint; then $\psi_a'''(0) = \psi_a'(0) + 8\Phi_a'(0) = -1/2$. The raw third source derivative lies in $(-40.00, -39.98)$, so positive scaling does not restore source reflection.

The enclosure calculation uses validated integration on $[0, 1]$, together with the complete tail bound, for $k \leq 3$,

$$\int_1^\infty u^k e^{u^2/4} e^{\lambda_n} \phi_n(u)\, du \leq \lambda_n \exp\{\lambda_n - 4 - (\lambda_n/2)e^4\}.$$

Indeed $k \log u + u^2/4 + 9u \leq (\lambda_n/2)e^{4u}$ for $u \geq 1$; its elementary endpoint constant is also enclosed. Centered jets and absolute moments extend the bounds across all of I. Fifty assertions at 160-bit precision and an independent replay support these finite claims; no sampled near-zero is substituted for root existence. □

The restriction to finite support is inessential. Add a sufficiently small positive multiple of $\sum_{n \geq 4} \phi_n$, and solve $\Phi_a'(0) = 0$ for the third weight. After this elimination, the Jacobian of $(H_a, H_{a,x})$ with respect to the remaining weight parameter and x is a nonzero multiple of $B(x_*) H_{a,xx}(T, x_*)$. The implicit function theorem preserves the collision and endpoint, with every weight positive. Every finite ordinary and adjacent-square localizing matrix is then strictly positive definite. This continuation is qualitative: no numerical perturbation radius is asserted. All resulting weights and normalizations remain independent of physical heat time.

On the control's full stationary predecessor, $\mathcal{L}_0/E = -1/(T-t) + O(1)$; its short admissible chords also become negative. Thus the relaxed assumptions cannot supply the required uniform signed estimates. The control retains periodicity in z and period two in τ, and obeys Jacobi heat, but fails the full elliptic multiplier. Its evenness in the spectral variable x does not imply evenness of its source in u.

9.2 Complete reflection fixes the coefficients

Theorem 9.2 (Weighted-comb reflection rigidity)**.** *Let $(a_n)_{n\in\mathbb{Z}}$ be an even complex sequence with $|a_n| \le C(1+|n|)^M$ for some $C > 0$ and finite $M \ge 0$. Define $F_a(q) = \sum_{n\in\mathbb{Z}} a_n e^{-\pi n^2 q}$ for $q > 0$. If*

$$F_a(q) = q^{-1/2} F_a(1/q) \qquad (q > 0), \tag{9.2}$$

then $a_n = a_0$ for every integer n. No coefficient positivity is needed.

Proof. Use $\widehat{f}(\xi) = \int f(x) e^{-2\pi i x\xi}\, dx$ and the even tempered comb $\mu = \sum a_n \delta_n$. For $g_q(x) = e^{-\pi q x^2}$, $\widehat{g}_q = q^{-1/2} g_{1/q}$, so (9.2) gives $\langle \nu, g_q\rangle = 0$, where $\nu = \mu - \widehat{\mu}$. Differentiation at $q = 1$ in the Schwartz topology annihilates every even Gaussian moment of ν; odd moments vanish by evenness. The function

$$\mathcal{G}(z) = \langle \nu, e^{-\pi x^2 + zx}\rangle$$

is entire, since the test functions and their parameter derivatives have uniform Schwartz seminorm bounds on compact z-sets. Every derivative at zero vanishes, hence $\mathcal{G} = 0$. On the imaginary axis this represents the Fourier transform of $e^{-\pi x^2}\nu$; Fourier injectivity and local division by the nowhere-zero Gaussian give $\nu = 0$. Thus $\mu = \widehat{\mu}$. Integer support makes $\widehat{\mu}$ period one as a distribution. Therefore μ is period one, and tests supported near successive integers give $a_{n+1} = a_n$. This proves the theorem; [15] records the full scope and defect identity. □

Corollary 9.3 (Source reflection and the primitive endpoint)**.** *In the class of Theorem 9.2, take $a_0 = 1$ and set, on the entire real u axis,*

$$P_a(u) = e^u F_a(e^{4u}) = e^u + 2\psi_a(u), \qquad \Phi_a = (P_a'' - P_a)/16.$$

If $\Phi_a(-u) = \Phi_a(u)$ and $\psi_a'(0) = -1/2$, then $a_n = 1$ for all n.

Proof. Local uniform convergence makes these functions real analytic. The difference $R(u) = P_a(u) - P_a(-u)$ satisfies $R'' - R = 0$, $R(0) = 0$, and $R'(0) = 2[1 + 2\psi_a'(0)] = 0$. Hence P_a is even, which is exactly (9.2); apply the theorem. □

Together with the normalized primitive endpoint, the complete odd source endpoint hierarchy likewise gives reflection by real analyticity. Proposition 9.1 retains only its first condition. Alternatively, imposing the full elliptic law on $\sum_n a_n e^{\pi i n^2\tau + 2\pi i n z}$ directly forces $a_{n+1} = a_n$ by reindexing. These relations supply coefficient coupling that support-only positivity omits. They identify the theta input, but an additional argument must still transfer that coupling to a favorable sign of the physical chord or stationary observation.

10 The remaining arithmetic-to-physical estimate

The strongest open target is now a specific two-point inequality, not an additional geometric realization. On a neighborhood of every candidate event, for both $j = 0, 1$, seek a source-prescribed estimate

$$\mathcal{C}_j(t; x, y) \ge -K E(t, x)^\gamma, \qquad \gamma > \tfrac{1}{2}, \tag{10.1}$$

with locally bounded K, a fixed positive allowed interval length, full stationarity at x, and h_j zero-free on $[x, y]$. The fixed-endpoint theorem proves the transfer to collision exclusion; the complete theta source has not been shown to satisfy this estimate. Its quantifiers include the fixed nearby endpoints used by the proof, not only a single shrinking window. The original one-sided turning modulus and stationary Laguerre bounds remain alternative sufficient targets, with their own domains.

10.1 Coefficient coupling with a complete remainder

The square-support control and reflection theorem distinguish two arithmetical inputs. Even all support-only localizers leave the weights free and permit a collision. Full modular reflection fixes the weights, but uniqueness alone says nothing about the signed physical transform. A useful next identity must use that coupling inside the complete bilocal chord, preserve full stationarity, and retain both source variables and their affine lower endpoints. The theta line-bundle translation law is another exact way to express the same coefficient coupling; scalar period two alone does not impose it.

A concrete candidate is an all-integer-cutoff family

$$\mathcal{C}_j \geq -AE(t,x)e^{aN^2} - Be^{-cN^2}, \qquad a \leq a_* < c_0 \leq c,$$

with source-derived locally uniform constants and cutoff threshold. The strict rate gap would give an error $o(\sqrt{E(t,x)})$ after optimization. Sub-Gaussian cutoff growth is also sufficient. Neither the complete decomposition nor its rate gap is established. The cutoff N is distinct from the unbounded spectral coordinate x; favorable cutoff scaling does not supply spectral coverage.

The transverse jet recurrence provides exact differential relations available for such an identity. The path-restricted ASD curvature does not itself provide a positive term, and an auxiliary norm still needs a proved comparison to the physical observation before it can pay a remainder.

10.2 A tested contour route and its limitation

The period-two contour deformation in [13] gives an exact identity $h_j = V_j + B_j$ with the shifted ray and bottom edge kept together. The leading exponential terms of the two sectors cancel; the bilocal observation accordingly contains all four sector products. Separate absolute sector bounds have exponential cost in spectral height and cannot provide a polynomial-in-height payment against bounded physical energy. This rejects that particular estimate, not the complete contour identity or a coefficient allowed to be merely locally bounded at each finite height. Detailed asymptotics and validated samples remain in the note.

Other proposals, including the natural fiber-norm lift, unchanged compact ASD completion, and global Pick-type positivity, are assessed in the numbered research overview and its linked strategy note [16, 17]. They are not additional premises of the theorems here. Their next useful contribution would be a new cancellation-preserving identity reaching (10.1), with a complete remainder.

10.3 Validation and coverage

A proposed coefficient rule should first be tested against the asymmetric heat-polynomial controls and the validated weighted-square collision. This identifies which extra arithmetic hypothesis is doing work. A theta pilot must then evaluate the complete signed observation, full stationary root, interval regularity, endpoints and tails using coefficients fixed before the calculation. A finite success motivates an analytic extension; it does not prove an all-cutoff estimate or coverage of the unbounded spectral line and the small-time limit. Those requirements, and the imported global hypotheses in Remark 2.2, remain indispensable for an RH conclusion.

Acknowledgments and research status

This research manuscript was prepared for Edward Baker with substantial assistance from OpenAI language models. The coordinating session used `gpt-6-astra` with reasoning effort `ultra`; internal

collaborators inherited that configuration. LLM assistance was used for drafting, derivations, literature checks, exact-arithmetic programs and internal reviews. The accompanying reviews distinguish analytic arguments, validated interval computations, and finite algebra checks. These internal checks are not external mathematical refereeing or formal verification. Edward B. Baker III's 2014 quaternion-form notes are acknowledged as exploratory motivation. The author and future reviewers retain responsibility for assessing the claims and their applicability.

A Verification scope and provenance

The verification package distinguishes analytic arguments from finite calculations. Seven exact checkers cover the geometry, Chern/ASD identities, complete signed bridge and heat profiles, with 1,230 overlapping assertions. Two further exact checkers add 5,667 assertions for the chord controls and reflection algebra. The weighted-square certificate and contour checker add 123 validated or mixed assertions using Arb/Acb with explicit complete tails. These totals are not counts of distinct theorems. The reflection-rigidity and arbitrary-multiplicity arguments are analytic proofs, not consequences of finite samples.

The chord replay uses 24 asymmetric or sign-reversed heat germs, 144 full stationary roots, and 1,152 zero-free intervals at three scales. Early intervals need not show the eventual negative sign; the asymptotic theorem is not replaced by universal finite-step negativity. The weighted-square calculation certifies a non-theta collision. The contour samples check complete identities and are not stationary theta sign certificates.

The reproducible scripts, output receipts and source hashes are indexed in the connection and profile manifest and arithmetic, contour and chord manifest. The integration review records the manuscript-specific proof and layout checks; the build record identifies this source and its exported PDF. Older manifests describe their own research checkpoints, including the then-unchanged initial manuscript. They are not silently rewritten as current build records.

The foundational material comes from Investigation 21 [1] and its Notes 56–63. Principal geometric arguments are restated, while the global theta-to-RH bridge retains its imported status. The present integration adds the stronger chord criterion, complete source bridge, metric classification and arithmetic discrimination from the linked Investigation 22 notes. The period-two experiment is summarized; less developed proposals remain in [16, 17]. Local research links are relative to the repository reading copy in `output/pdf`; the source remains the stable editable manuscript, with milestones indexed in `DRAFT_HISTORY.md` and no new snapshot folders.

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